

Fig. 3: Actual-size PCB layout of an automatic evening lamp

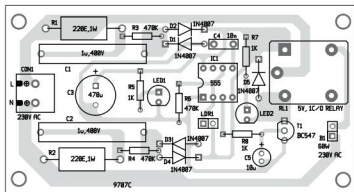


Fig. 4: Component layout of the PCB

PARTS LIST

Semiconductors:

- IC1 - 555 timer
- T1 - BC547 npn transistor
- D1-D5 - 1N4007 rectifier diode
- LED1, LED2 - 5mm LED
- LDR1 - Small-size LDR

Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):

- R1, R2 - 220-ohm, 1 watt
- R3, R4, R6 - 470-kilo-ohm
- R5, R7, R8 - 1-kilo-ohm

Capacitors:

- C1, C2 - 1 μ F, 400V polyester, X-rated
- C3 - 470 μ F, 25V electrolytic
- C4 - 10nF ceramic disk
- C5 - 10 μ F, 25V electrolytic

Miscellaneous:

- CON1 - 2-pin terminal connector
- RL1 - 5V, 1C/O relay
- B1 - 60W, 230V bulb
- 8-pin IC base

operation of the relay. Freewheeling diode D5 eliminates back EMF from the relay coil and protects T1 during its switch off. Red LED2 indicates the actuation of the relay.

Construction and testing

An actual-size, single-side PCB layout for the automatic evening lamp is shown in Fig. 3 and its component lay-

out in Fig. 4. After assembling the circuit on a PCB, enclose it in a suitable plastic case.

Give sufficient spacing between the power supply section and the remaining circuit. Provide holes on the front side of the enclosure for LEDs and LDR. Connect phase line (L) to the common contacts of the relay and neutral line (N) for the bulb to the N/O (normally open) contacts of the relay. A 5V PCB relay is

used. Ratings of the relay must match with the load. Since the circuit is directly powered from high-volt AC, extreme care is necessary during testing.

First assemble the power supply section up to green LED and connect to AC lines. If the green LED turns on, power supply section is alright. After disconnecting the circuit from mains, assemble the circuit around IC1. Test this part using a 9V battery connected across capacitor C3. If relay RL1 energises after masking LDR1, the bi-stable section is working.

Now the relay connections can be done. Keep the unit outdoor in a place where sufficient light is available. Light from the lamp should not fall on LDR1.

Caution. Since this circuit has mains voltage on board, extreme precautions need to be taken. Do not troubleshoot when it is connected to the mains. Test only after taking adequate precautions to prevent shock hazards. ●

The author is a lecturer in a government women college in Kerala

Limited Time Offer

12 eZine
Editions of
Electronics
For You
(MRP ~~₹600~~)

only ₹100



ElectronicsForYou
eZine
Smart way of online reading

The e-zines have the same content-editorial and advertisements as the print editions. But, they are environment friendly and easier to archive and refer to. All you need is an Internet-enabled device with a decent bandwidth.

To Buy online
offers.kitsNspares.com

+91-88001 00761 | +91-11-26371661

info@kitsNspares.com

KitsNspares.com
For the 'engineer' in you